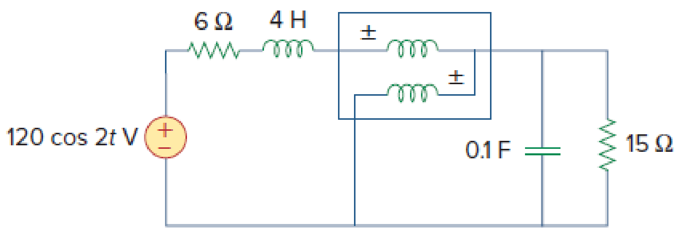
Chapter 11, Problem 77P

Problem

What is the reading of the wattmeter in the network of Fig. 11.92?

Figure 11.92:



Step-by-step solution

Step 1 of 4

The wattmeter measures the power absorbed by the parallel combination of 0.1 F&150 Ω.

$$120\cos(2t) \Rightarrow 120\angle 0^\circ$$

$$\omega = 2 \text{ rad/sec}$$

$$4\text{H} = j\omega L$$

$$4\text{H} = j8\ \Omega$$

$$0.1\text{F} = \frac{1}{j\omega C}$$

$$0.1\text{F} = -j5\ \Omega$$

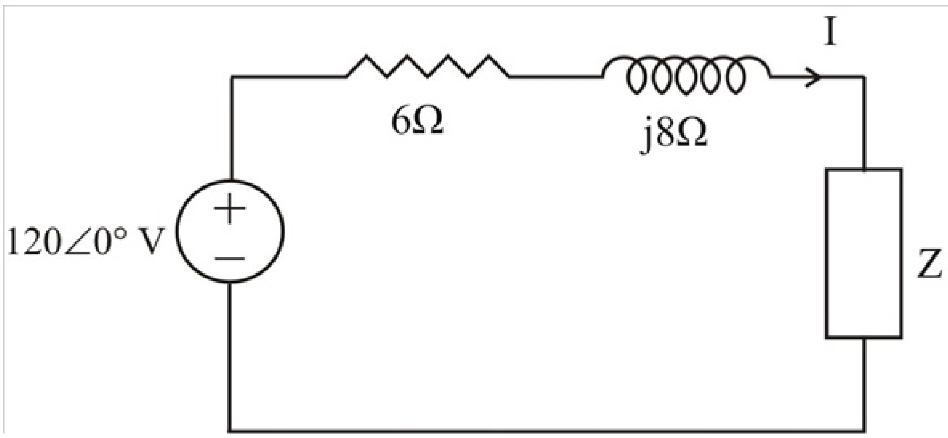
$$Z = 15\Omega \| \ (-j5\Omega)$$

$$Z = \frac{15(-j5)}{15-j5}$$

$$Z = 1.5 - j4.5\ \ \Omega$$

Step 2 of 4

Consider the following circuit,



Step 3 of 4

$$I=\frac{120}{(6+j8)+(1.5-j4.5)}$$

$$I=\frac{120}{7.5+j3.5}$$

$$I=14.5\angle-25.02^\circ\ \text{A}$$

Step 4 of 4

$$S=\frac{1}{2}V\times I^*$$

$$S=\frac{1}{2}|I|^2\times Z$$

$$S=\frac{1}{2}\times(14.5)^2\times(1.5-j4.5)$$

$$S=157.69-j473.06$$

The wattmeter reads,

$$P=\text{Re}(S)=\boxed{157.69\ \text{W}}$$